Unusual Mobility Behaviour Detection for Security Surveillance

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The Surveillance Challenge

Data Overload

Modern CCTV networks generate petabytes of footage daily, far exceeding human review capacity. Most anomalous events go unnoticed in real time.

Rule-Based Limitations

Traditional systems encode only anticipated behaviours. They fail with novel patterns, lighting changes, and varying crowd densities.

Labelling Problem

Supervised deep learning requires labelled anomaly data, but anomalies are rare and expensive to collect ethically.

Our Solution: Unsupervised Anomaly Detection

01

Learn Normal Behaviour

System learns representation of normal movement from unlabelled footage without anomaly examples

02

Extract Dual Features

Combines geometric trajectory analysis with deep video embeddings for comprehensive understanding

03

Score Anomalies

Measures distance from learned normality to produce continuous severity scores

04

Generalise Across Environments

No labelled data required at deployment, enabling natural generalisation across scenes

System Architecture

Video Sampling

5 FPS, 640px width

YOLOv8n Detection

Confidence ≥0.4

DeepSORT Tracking

Stable identities

Feature Fusion

61D vectors

HDBSCAN Clustering

Anomaly scores

Branch A: Trajectory Features

11-Dimensional Geometric Analysis

- Mean speed and acceleration patterns
- Path efficiency and directional variance
- Stop ratio and loitering detection
- Total distance versus net displacement

Captures movement geometry without visual appearance

Branch B: VideoMAE Embeddings

VideoMAE Large Model

OPear/videomae-large-finetuned-UCF-Crime domain-adapted embeddings

1024-Dimensional Vectors

CLS token extraction from 16-frame clips, mean-pooled per track

Surveillance-Optimised

Encodes camera artifacts, crowd occlusion, low resolution patterns

Feature Fusion Strategy

Independent Normalisation

Separate StandardScaler for each branch, fitted only on normal training data

PCA Dimensionality Reduction

Branch B reduced from 1024 to 50 components (85%+ variance retained)

Weighted Concatenation

0.3 × Trajectory + 0.7 × VideoMAE produces 61D fused vectors

Why 0.3/0.7 split? Domain-adapted video embeddings show greater discriminative power than geometric features in ablation studies

Performance Results

0.82

Frame-Level AUC

Exceeds 0.80 target, competitive with weakly-supervised methods

0.78

Recall @ 0.75

High detection rate at operational threshold

0.71

Precision @ 0.75

Low false alarm rate for operator efficiency

6

AUC Gain

Improvement from single branch to fused operation

Key Findings and Insights

Dual-Branch Superiority

6-point AUC improvement validates trajectory and appearance capture complementary anomaly dimensions

Continuous Scoring Matters

Centroid-distance scores yield 0.82 AUC vs 0.67 for binary HDBSCAN labels alone

Domain Adaptation Critical

UCF-Crime fine-tuned VideoMAE outperforms generic Kinetics models substantially

Engineering Pitfalls

numpy 2.x breaks filterpy, VideoMAEForVideoClassification discards CLS tokens

Future Research Directions

Real-Time Streaming

VideoMAE quantisation and TensorRT optimisation for live alerting instead of post-processing

Multi-Camera Tracking

Person re-identification across camera handoffs with consistent identity maintenance

Online Learning

Adapt normal behaviour distribution to gradual scene changes without full retraining

Beyond Persons

Abandoned objects, vehicle intrusions, crowd density anomalies for broader utility